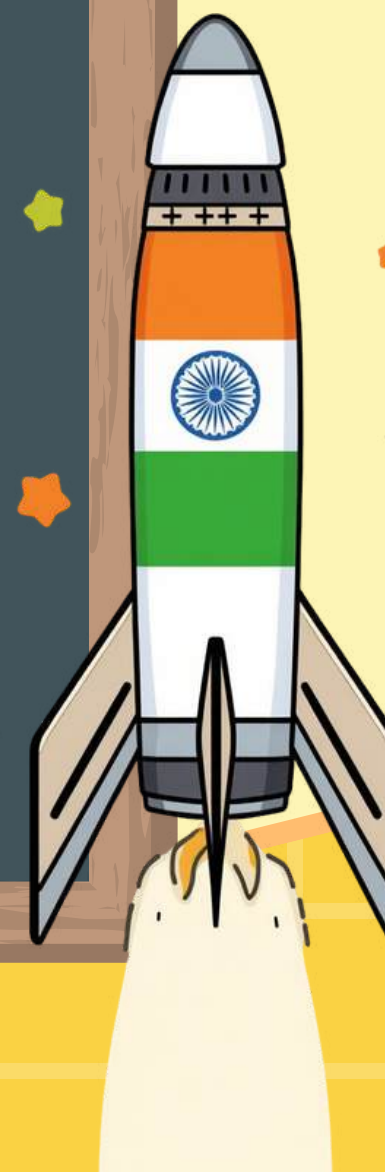
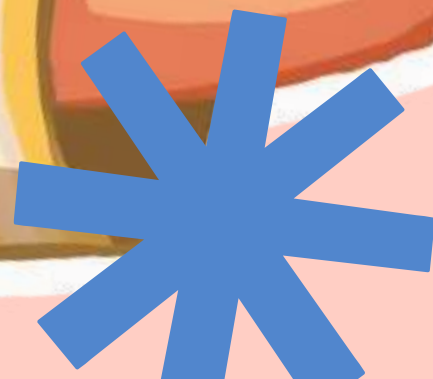


ROCKETS — NEWTON'S 3RD LAW!





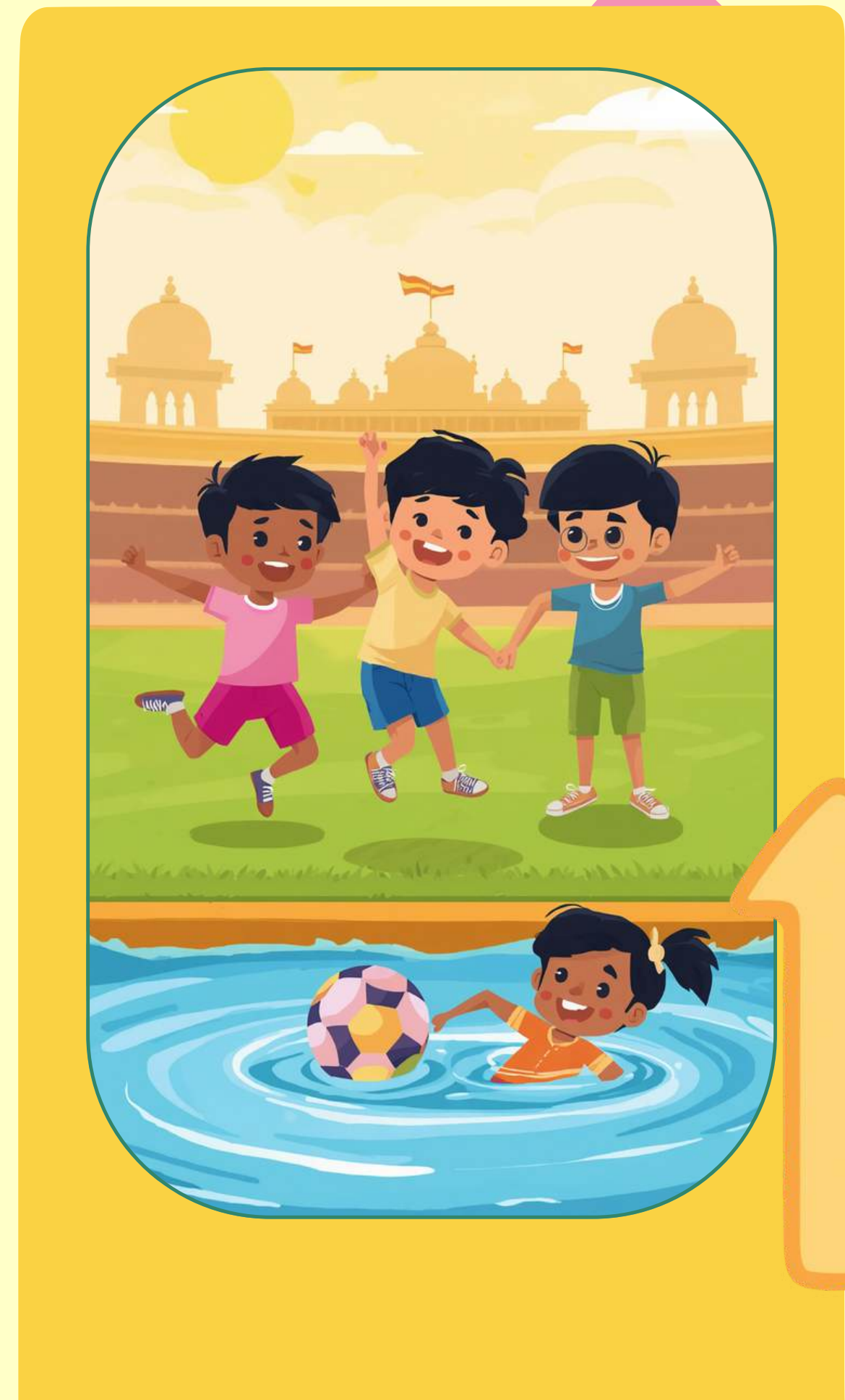
How does a rocket go
UP if there's nothing
to push against in
space?



Push & Push-Back!

Every push has a push back! When you jump, you push the ground DOWN and the ground pushes YOU up! When you swim, you push water BACK and you move FORWARD!

Action creates reaction!





Newton's 3rd Law of Motion

When you push something, it pushes YOU back – equally hard, in the opposite direction! This is called Action and Reaction!

Push = Push Back!





3RD LAW

More Indian Examples!

Newton's 3rd Law is everywhere around us! Let's see how it works in our daily Indian life.

Push back = Move forward!

Let's Make a Balloon Rocket!

Materials you'll need:

- Long balloon
- String (3-5 metres)
- Plastic straw
- Sticky tape
- 2 chairs

Gather all materials
before we start!



Build Your Balloon Rocket!

1. Thread string through straw
2. Tie string tightly between chairs
3. Blow up balloon (don't tie!)
4. Tape balloon to straw

Follow these easy steps!



Ready? 3... 2... 1... BLAST OFF!

Let go of the balloon and watch it zoom along the string! See how fast it flies across the room!

Release and watch the magic!





BACKET

LEFT

air →
rush out

What Happened?

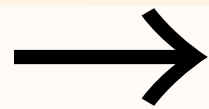
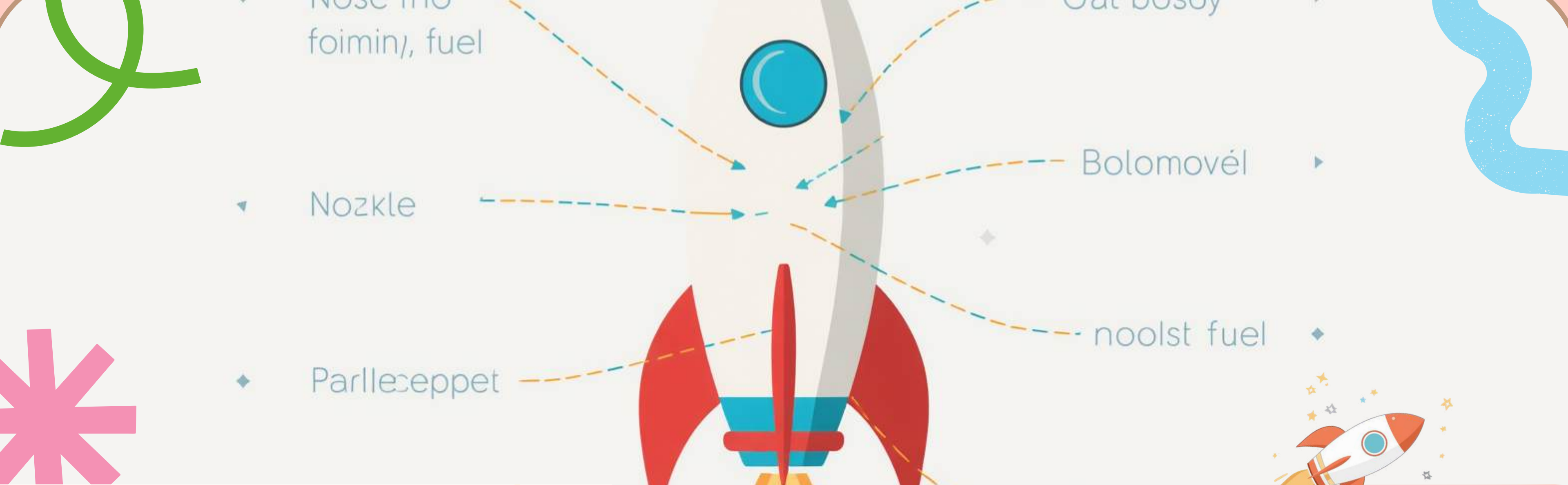
Air rushes OUT the back → WHOOOOSH!

Balloon zooms FORWARD → ZOOM!

Air goes BACK → Balloon goes FRONT!

This is Newton's 3rd Law!

Action creates reaction!



NOZZLE



How a Real Rocket Works

Gas shoots DOWN → Rocket goes UP! Just like our balloon experiment!

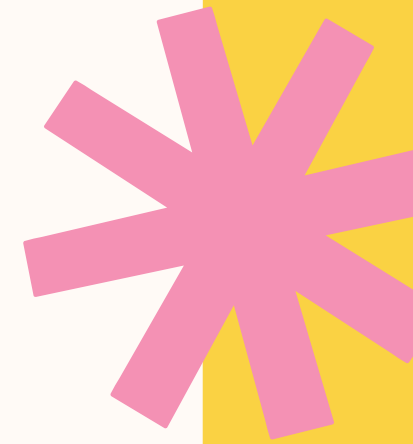
Nose · Body · Nozzle = Rocket Parts!

ISRO Pride – India's Space Champions!

ISRO = Indian Space Research Organisation

- PSLV: 104 satellites in ONE launch! (World record!)
- Chandrayaan-3: Landed on Moon's south pole! (2023)
- Mangalyaan: Reached Mars first try! (2014)
- Dr. A.P.J. Abdul Kalam: Missile Man of India

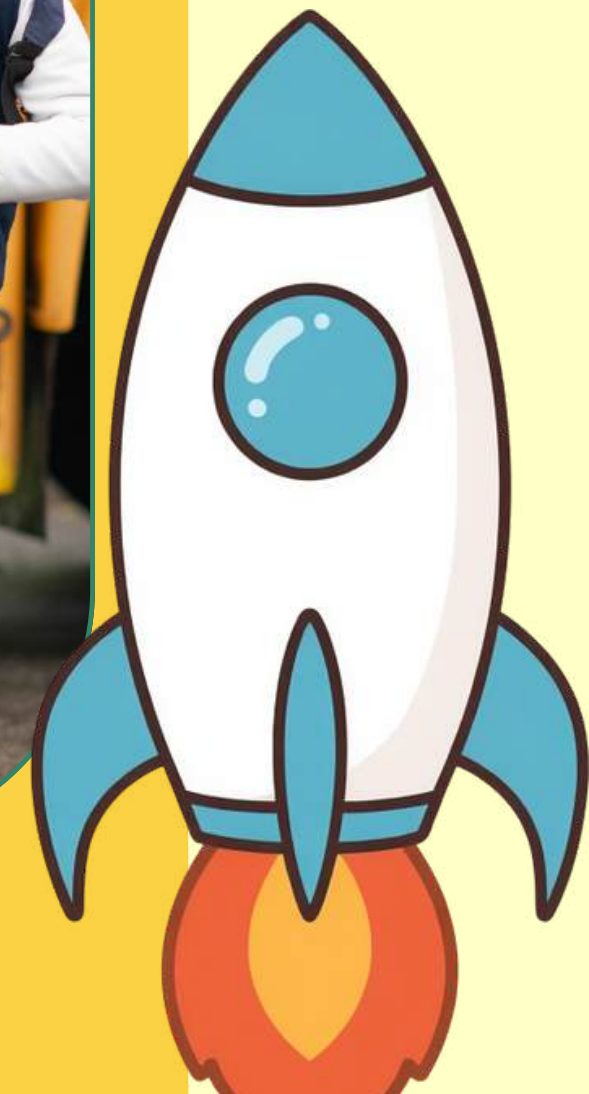
Dream big like ISRO!



Rocket Riddles!

- Rocket pushes gas DOWN. Which way does rocket go? → UP!
- Boatman pushes water BACK. Which way does boat go? → FORWARD!
- What is this law called? → Newton's 3rd Law!
- Which Indian space agency uses this? → ISRO!

Test your knowledge!
Raise your hand!



**DREAM BIG!
YOU ARE A
FUTURE ISRO
SCIENTIST!**

 JAI HIND! JAI VIGYAN! 



JAI HIND! JAI VIGYAN!

Dream big! You are a future ISRO scientist!

